

Kaggle Heart Disease Classification

Interpretable binary classification with validation and calibration

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1 Abstract

This article presents the final project in the current portfolio roadmap: a Kaggle heart disease classification workflow. The dataset contains 303 records and a heart disease prevalence of 54.5%. The workflow compares logistic regression and decision tree models with stratified validation, ROC-AUC, confusion metrics, calibration bins, and standardized coefficient interpretation. The selected model is logistic, with holdout ROC-AUC of 0.863, accuracy of 80.3%, sensitivity of 87.9%, and specificity of 71.4%. This is a portfolio benchmark, not a clinical decision tool.

2 Introduction

Heart disease classification is a classic structured-data benchmark. It is useful for a portfolio because it requires the analyst to handle categorical-style numerical variables, evaluate binary classification metrics, and communicate model limits.

The research question is:

Which variables help classify heart disease presence,
and how can the model remain interpretable?

3 Data

Table 1: Dataset summary.

Item	Value
Canonical source	Kaggle Heart Disease Dataset
Records	303
Predictors	13
Outcome prevalence	54.5%
Target	Heart disease presence

4 Methods

The workflow uses a stratified 80/20 holdout split and five-fold stratified cross-validation on the training set. Logistic regression is trained on standardized predictors, while the decision tree uses the original numeric feature scale. ROC-AUC is computed manually, and confusion metrics are reported at threshold 0.50.

5 Results

5.1 Holdout Metrics

Table 2: Holdout model metrics at threshold 0.50.

Model	Accuracy	Sensitivity	Specificity	Precision	F1	ROC-AUC
logistic	80.3%	87.9%	71.4%	78.4%	82.9%	0.863
tree	73.8%	69.7%	78.6%	79.3%	74.2%	0.746

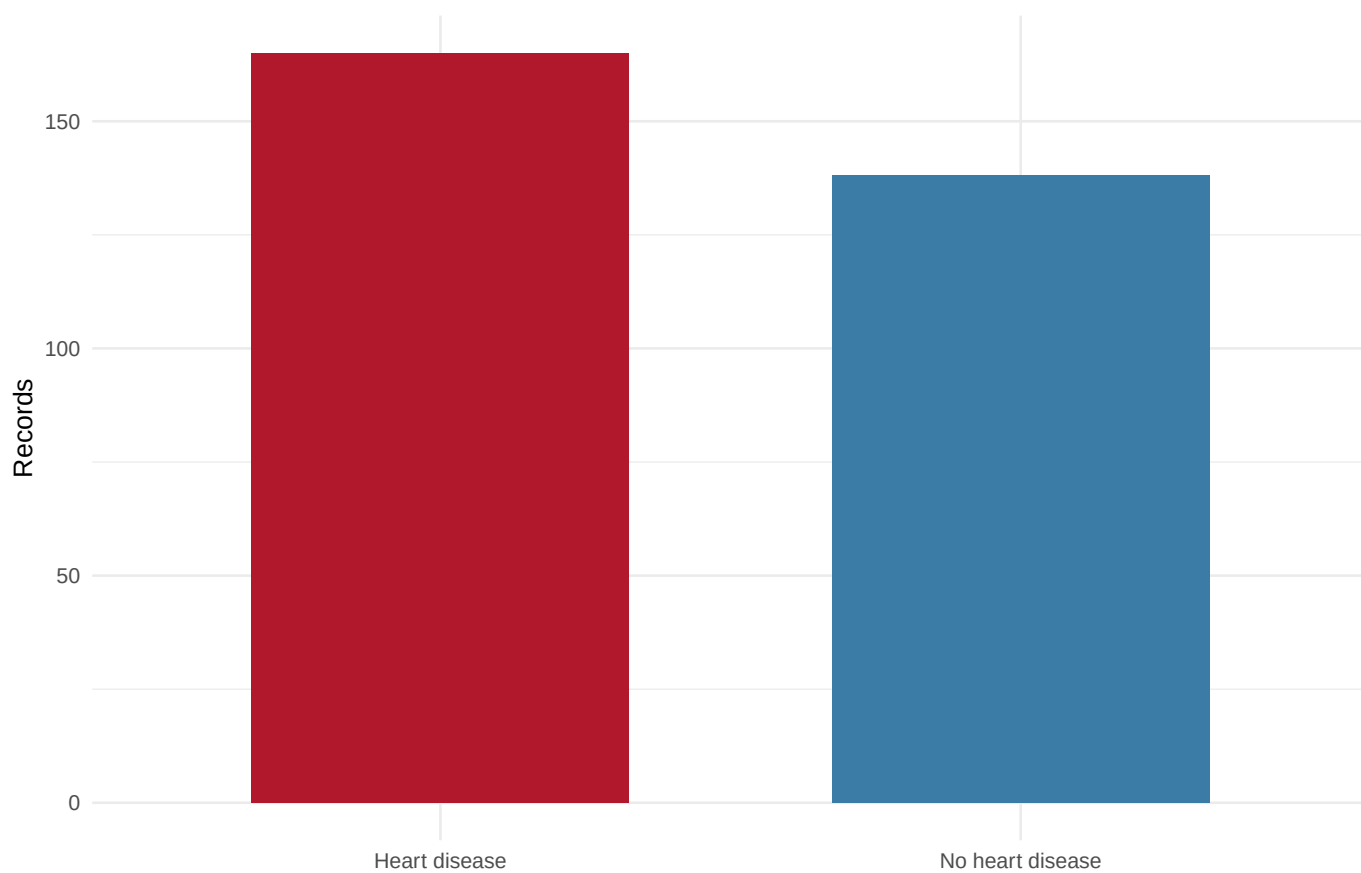


Figure 1: Outcome distribution.

5.2 Confusion Matrix

Table 3: Selected-model holdout confusion matrix.

Observed	Predicted	Records
No heart disease	No heart disease	20
No heart disease	Heart disease	8
Heart disease	No heart disease	4
Heart disease	Heart disease	29

5.3 ROC Curve

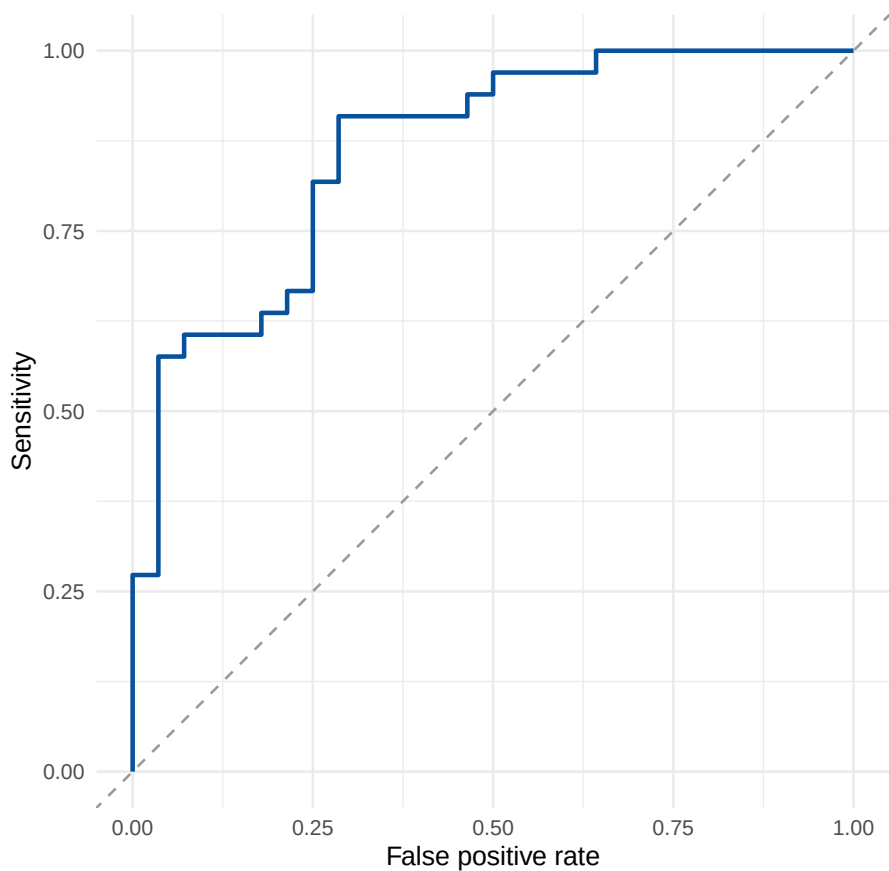


Figure 2: Selected-model ROC curve.

5.4 Cross-Validation

Table 4: Five-fold cross-validation summary.

Model	Accuracy	Sensitivity	Specificity	F1	Mean ROC-AUC
logistic	82.2%	87.2%	76.4%	84.3%	0.906
tree	79.8%	80.4%	79.1%	81.2%	0.829

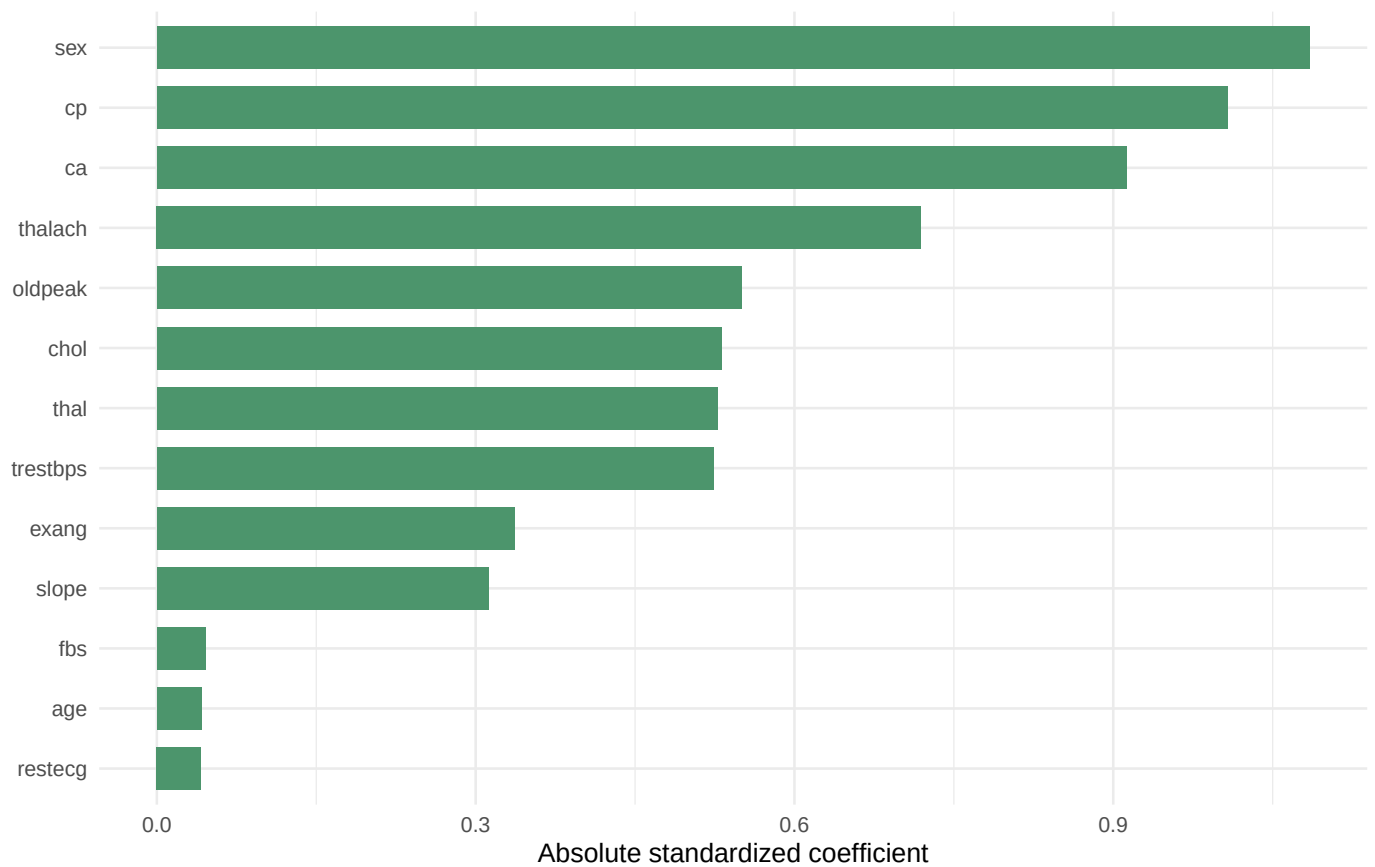


Figure 3: Logistic regression variable importance.

5.5 Variable Interpretation

Table 5: Standardized logistic regression coefficients.

Variable	Coefficient	Direction
sex	-1.085	Lower probability
cp	1.007	Higher probability
ca	-0.913	Lower probability
thalach	0.719	Higher probability
oldpeak	-0.550	Lower probability
chol	-0.532	Lower probability
thal	-0.527	Lower probability
trestbps	-0.524	Lower probability
exang	-0.337	Lower probability
slope	0.312	Higher probability
fbs	-0.046	Lower probability
age	-0.042	Lower probability
restecg	0.041	Higher probability

5.6 Calibration

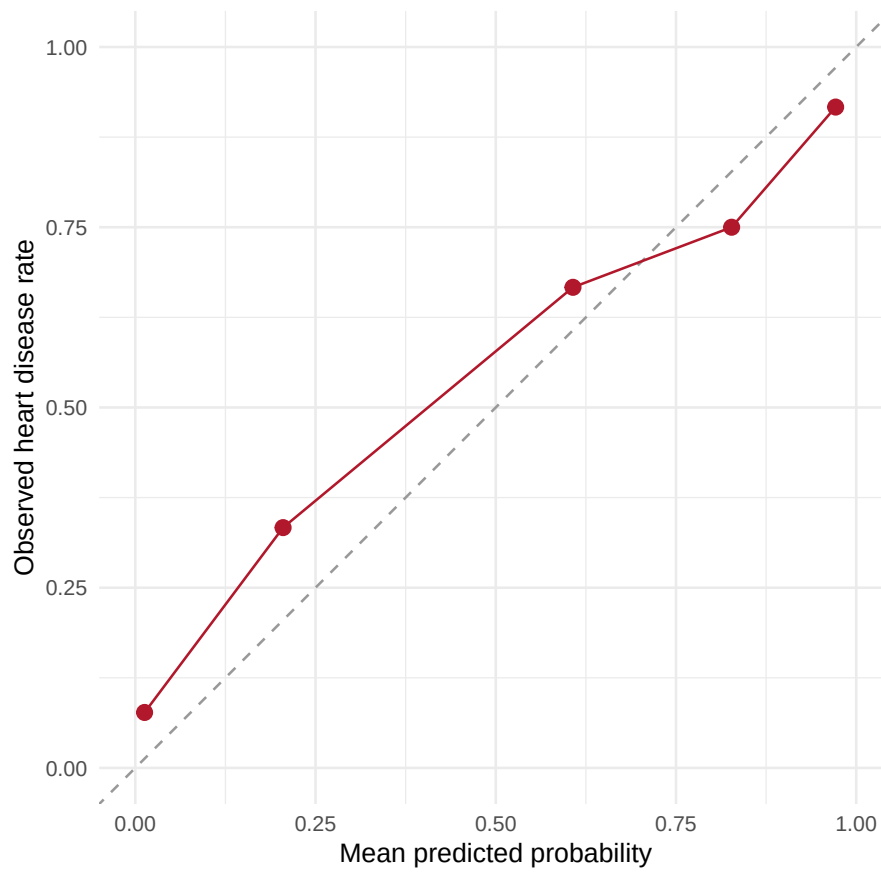


Figure 4: Calibration bins for the selected model.

6 Discussion

The selected model provides an interpretable classification benchmark. Logistic regression coefficients show which standardized predictors are associated with higher or lower predicted probability, while the decision tree provides a simple nonlinear comparison.

The model should be read as a teaching and portfolio artifact. In clinical settings, threshold choice would depend on screening goals, harm tradeoffs, prevalence, and external validation.

7 Limitations

The dataset is small, benchmark-oriented, and not sufficient for medical deployment. It lacks longitudinal clinical history, medications, symptoms, diagnostic imaging, care pathways, and representative sampling. The output is not medical advice, diagnosis, triage, or a clinical recommendation.

8 Conclusion

S14 completes the current portfolio roadmap with a validated, interpretable heart disease classification project. The workflow includes model comparison, ROC-AUC, confusion metrics, calibration, variable interpretation, and careful clinical limitations.

9 References

Kaggle Heart Disease Dataset. <https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>

UCI Heart Disease Dataset. <https://archive.ics.uci.edu/dataset/45/heart+disease>

Public CSV mirror used for local reproducibility. <https://raw.githubusercontent.com/kb22/Heart-Disease-Prediction/master/dataset.csv>